1. A Y-connected balanced three-phase generator with an impedance of 0.4+j0.3 Ω per phase is connected to a Y-connected balanced load with an impedance of 24+j19 Ω per phase. The line joining the generator and the load has an impedance of 0.6+j0.7 Ω per phase. Assuming a positive sequence for the source voltages and that $V_{an} = 120\angle 30°\,V$. Find: (a) the line voltages, (b) the line current.

2. One line voltage of a balanced Y-connected source is $V_{AB} = 120\angle -20°\,V$ If the source is connected to a ▲-connected load of $20\angle 40°\,\Omega$ find the phase and line currents. Assume the abc sequence.

3. A positive-sequence (abc), balanced ▲- connected source supplies a balanced ▲-connected load. If the impedance per phase of the load is 18+j12 Ω and $I_a = 9.609\angle 35°\,A$ find I_{AB} and V_{AB}.

4. In a balanced ▲ – Y circuit, $V_{ab} = 240\angle 15°\,V$ and $Z_Y = 12 + j15\,\Omega$. Calculate the line currents.

5. Calculate the line current required for a 30-kW three-phase motor having a power factor of 0.85 lagging if it is connected to a balanced source with a line voltage of 440 V.

6. Find the line currents in the below three-phase circuit of Fig. 1 and the real power absorbed by the load.

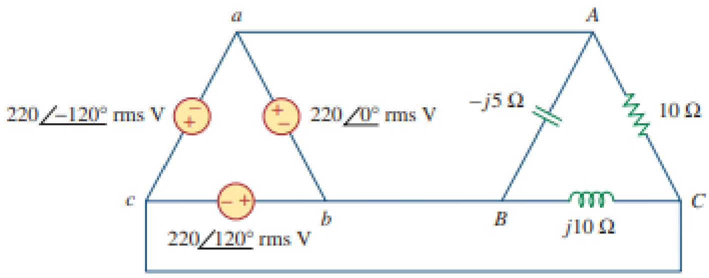

Fig. 1

7. A parallel resonant circuit has R = 100 kΩ, L = 20mH, and C = 5nF. Calculate quality factor, resonant frequency, bandwidth and cut-off frequencies.

8. Calculate the resonant frequency of the circuit in Fig. 2.

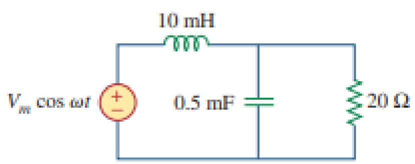

Fig. 2

9. In a series RLC circuit, If the current at resonance is 10 A, then what will be the value of current corresponding to half power delivered in the circuit.

10. A series RLC circuit resonates at 1.5 kHz and consumes 50 W from a 50 V AC source operating at the resonant frequency. If the bandwidth is 750 Hz then what will the values of circuit elements R, L and C.